

Stanley College of Engineering & Technology for Women (A)**B.E (CSE/CME/AI&DS/IT) IV Semester Main Examination -July-2023****Operating Systems****Time: 03 Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. List out different services of operating systems. [2M CO1 BTL1]
- b. What is a thread? Name the benefits of thread. [2M CO2 BTL1]
- c. Classify the necessary conditions for Deadlock. [2M CO3 BTL2]
- d. What is virtual memory? [2M CO4 BTL1]
- e. Describe page fault. [2M CO4 BTL2]
- f. Mention the key evaluation points of RAID System. [2M CO5 BTL1]

Part-B**4X12=48M**

2. a) Define OS and explain functions of OS. [6M CO1 BTL2]
- b) What is system call? Identify various types of system calls in detail. [6M CO1 BTL3]
3. a) From the below given data. Draw Gantt charts illustrating the execution of the processes using FCFS and SJF scheduling algorithms and calculate the average turnaround time and waiting time for them. [6M CO2 BTL2]

Processes	Arrival Time	Burst Time
P1	0	4
P2	1	2
P3	2	2
P4	3	5
P5	4	6

- b) Define process. Draw and explain various Process States in detail. [6M CO2 BTL3]
4. a) Illustrate about critical-section problem and Peterson's solution in concurrency. [6M CO3 BTL2]
- b) How to apply the monitors to solve the dining philosopher's problem with example? [6M CO3 BTL3]
5. a) Differentiate external fragmentation with internal fragmentation. [6M CO4 BTL4]
- b) Distinguish about Logical and Physical Addressing. [6M CO4 BTL4]
6. Consider the following page reference string:
 7,2,3,1,2,5,3,4,6,7,7,1,0,5,4,6,2,3,0,1
 Assuming the demand paging with three frames, determine how many page faults would occur for the following replacements algorithms?
 a. LRU b. FIFO c. Optimal [12M CO4 BTL5]
7. a) Summarize about file attributes, file operations, and file types. [6M CO5 BTL2]
- b) Discuss any two of disk scheduling algorithms with reference blocks. [6M CO5 BTL6]

Hall Ticket No. -

Code: R123752

Stanley College of Engineering & Technology for Women (A)

B.E (CSE/CME) IV Semester (Main) Examinations-July-2023

Database Management Systems

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
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Part-A

6X2=12M

- 1.a. What is data model? [2M CO1 BTL1]
- b. Who is Database Administrator? [2M CO1 BTL1]
- c. What is join operation? [2M CO2 BTL1]
- d. What are entity set and relationship set? [2M CO3 BTL1]
- e. What is the main difference between sparse index and dense index? [2M CO4 BTL1]
- f. Define dead lock. [2M CO5 BTL1]

Part-B

4X12=48M

2. a) Explain briefly, the applications of DBMS. [6M CO1 BTL2]
b) List out the database users. Explain. [6M CO1 BTL2]
3. a) When not to use DBMS? [6M CO1 BTL2]
b) Interpret concepts of schema, instance and data independence. [6M CO1 BTL3]
4. a) Interpret integrity constraints with examples. [6M CO2 BTL3]
b) Differentiate between nested and correlated queries. [6M CO2 BTL3]
5. a) Draw the complete ER diagram for the following Bank database. [8M CO3 BTL3]
i) There are multiple banks and each bank has many branches and each branch has multiple customers.
ii) Customers have various types of accounts.
iii) Some customers also had taken different types of loans from these bank branches.
iv) One customer can have multiple accounts and loans.
b) Interpret 3NF with an example. [4M CO3 BTL3]
6. a) Explain storage structure. [6M CO4 BTL2]
b) Explain about serializability. [6M CO4 BTL2]
7. a) Interpret compatibility of locks. [6M CO5 BTL3]
b) Interpret lock based protocols. [6M CO5 BTL3]

Hall Ticket No. -

Code: R123757

Stanley College of Engineering & Technology for Women (A)

B.E (CSE/CME/EEE) IV Semester (Main) Examinations -July-2023

Effective Technical Communication

Time: 03 Hours

Max. Marks-60

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Part-A

6X2=12M

1. a. Explain any TWO barriers for communication. [2M CO1 BTL 1]
b. What is the process of communication? [2M CO2 BTL1]
c. Explain any TWO types of listening skills. [2M CO3 BTL1]
d. What is kinesics? [2M CO4 BTL2]
e. Write about the purpose of user manual. [2M CO5 BTL2]
f. Write advantages of an Email. [2M CO4 BTL3]

Part-B

4X12=48M

2. a) Explain various barriers for technical communication. [6M CO1 BTL2]
b) Discuss various techniques for effective presentation. [6M CO2 BTL3]
3. a) What are verbal and non-verbal communications? Mention some persuasive techniques. [6M CO2BTL3]
b) Write a user manual for a smart watch. [6M CO3 BTL3]
4. a) What is mobile etiquette? Explain in detail. [6M CO4 BTL1]
b) Define "Agenda" and "Minutes of meeting." Write the purpose of Agenda and Minutes of Meeting. [6M CO4 BTL3]
5. a) Imagine that you purchased an LG Top Load washing machine from online platform Amazon. You received a damaged product. Write a complaint letter to the Customer Care, Amazon. [6M CO4 BTL6]
b) Who writes and reads reports? Explain the elements of effective business reports. [6M CO4 BTL6]
6. a) You would like to open a startup business after your graduation. Prepare your business proposal for the same. [6M CO4 BTL6]
b) What is report? Write various types of a report in detail. [6M CO5 BTL2]
7. a) Explain the barriers for effective listening. [6M CO2 BTL2]
b) What does 'listening mean'? Explain Active and Passive listening and advantages of listening. [6M CO2 BTL2]

Stanley College of Engineering & Technology for Women (A)

B.E (CSE/CME) IV Semester (Main) Examinations-July-2023

Automata Theory Languages and Computation

Time: 03 Hours

Max. Marks-60

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Part-A

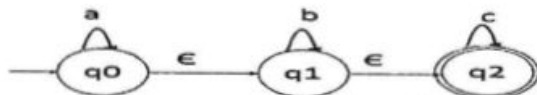
6X2=12M

1. a. List the applications of finite automata.

[2M CO1 L1]

- b. Define ϵ -closure and find the ϵ closures of all states.

[2M CO1 L1]



- c. What is Arden's theorem?

[2M CO2 L1]

- d. What is ambiguity in CFG? What are the different methods to remove ambiguity?

[2M CO3 L1]

- e. Define Pushdown automata.

[2M CO4 L1]

- f. State post correspondence problem.

[2M CO5 L1]

Part-B

4X12=48M

2. a) Identify the differences between DFA and NFA.

[4M CO1 L3]

- b) Convert NFA to DFA.

[8M CO1 L3]

	0	1
→ p	{p, q}	{p}
q	{r}	{r}
r	{s}	∅
*s	{s}	{s}

3. a) Construct a NFA equivalent to the Regular Expression $01^*00(0+1)^*$

[6M CO2 L3]

- b) Define pumping lemma and Show that the language $L = \{a^p : p \text{ is prime}\}$ is not regular.

[6M CO2 L3]

4. a) Define regular expression write the Regular expression and equivalent Finite automata for the following:

[6M CO2 L4]

- i. Set of all strings beginning with 1 and ending with 1

- ii. Set of all strings having three consecutive zeros

- b) Define CFG and construct a CFG for the following language.

[6M CO3 L3]

$$L = \{a^n b^m / \text{where } n \neq m\}$$

5. a) Construct LMD and RMD for the string 'aabbaa' from the grammar. [6M CO3 L3]
 $S \rightarrow aAS / a;$
 $A \rightarrow SbA / SS / ba$
- b) Convert the following CFG to Chomsky normal form. [6M CO3 L3]
 $S \rightarrow 0A | 1B$
 $A \rightarrow 0AA | 1S | 1$
 $B \rightarrow 1BB | 0S | 0$
6. a) Design a pushdown automata which accepts $L = \{wcw^r / w \in (a+b)^*\}$. [6M CO4 L3]
- b) Discuss Types of Turing Machines. [6M CO4 L3]
7. a) Discuss about Recursive and recursively enumerable languages. [6M CO5 L2]
- b) Write short notes on Linear Bounded Automata and context sensitive languages. [6M CO5 L3]

Hall Ticket No. -

Code: R123742

Stanley College of Engineering & Technology for Women (A)

B.E (CME) IV Semester (Main) Examinations-July-2023

Design and Analysis of Algorithms

Time: 03 Hours

Max. Marks-60

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Part-A

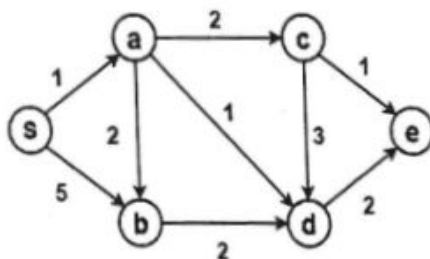
6X2=12M

- a. Rank the order of growth. Compare the order of growth $n!$ and 2^n . [2M CO1 BTL5]
b. Recall Assignment problem. [2M CO2 BTL1]
c. Distinguish prim's and Kruskal's algorithm. [2M CO3 BTL2]
d. Infer the convex hull problem. [2M CO3 BTL4]
e. Demonstrate the obstacles in constructing a minimum spanning tree. [2M CO3 BTL3]
f. Formulate the steps to search a pattern in the built suffix tree? [2M CO4 BTL6]

Part-B

4X12=48M

- a) List out the steps that need to design an algorithm. [6M CO3 BTL1]
b) How you will measure input size of algorithm. [6M CO1,3 BTL1]
- a) Define a Mathematical analysis of Non-recursive algorithms. [5M CO1 BTL4]
b) Tell about the efficiency of finding the element with maximum value in a given Array with the help of General plans. [7M CO1,3 BTL4]
- Write the algorithm of single source shortest path. Given a graph and a source vertex in the graph, find the shortest paths from the source to all vertices in the given graph.



[12M CO2,4 BTL5]

- Generalize the following instance of the 0/1, knapsack problem for given the knapsack capacity in $W=5$ using dynamic programming and explain it. [12M CO3,4 BTL6]

Item	Weight	Value
1	4	\$10
2	3	\$20
3	2	\$15
4	5	\$25

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6. a) Apply the backtracking concept to solve the 8-Queen problem. [6M CO4 BTL3]
b) Discover the procedures to solve travelling salesman problem. [6M CO4 BTL3]
7. a) Let $A = \{l/119, m/96, c/247, g/283, h/72, f/77, k/92, j/19\}$ be the letters and its frequency of distribution in a text file. Identify the suitable Huffman coding to compress the data. [6M CO4 BTL2]
b) Compare the P problem with NP problems. [6M CO3 BTL2]

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Part-B**4X12=48M**

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